

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY



CSIT (BSCS SPECIALIZATION IN CYBER SECURITY)

CYBER SECURITY LAWS & REGULATIONS (CT-376)

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PROJECT: OSINT MACHINE

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1. Introduction

In the modern digital landscape, cyber threats such as phishing, impersonation, and financial scams are rampant. Security teams are often presented with suspicious digital identities—emails, usernames, and URLs—that serve as the initial entry point for these attacks. Manually investigating these artifacts is timeconsuming and inefficient.

This report details the design and implementation of a specialized Open-Source Intelligence (OSINT) machine, built as a virtual machine (VM), to automate the forensic investigation of suspicious digital identities. Tailored for a specific use case, this OSINT machine integrates a curated suite of tools to provide a rapid, in-depth analysis of an email address, username, and associated web infrastructure from a single, controlled environment.

Use Case: Investigating a Suspicious Digital Identity (Email + Username + URL).

Scenario: A cybersecurity team receives a suspicious email from `primefundshelpdesk@gmail.com` signed by a user `primefunds\_support`, containing a link to `http://primefunds-login.com/verify`. The goal is to determine the legitimacy of this entity and assess potential risks.

2. Description of the OSINT Machine

Our OSINT machine is a Linux-based virtual machine pre-configured with a powerful, interoperable toolkit. It is designed to be a self-contained forensic workstation, ensuring analysis is conducted in an isolated environment to prevent accidental interaction with potentially malicious resources.

2.1 Architecture & Workflow

The machine follows a linear, logical workflow where the output of one tool can inform the use of the next. The architecture is modular, with each tool serving a distinct intelligence-gathering purpose.

Investigation Workflow:

Suspicious Email & Username → Mailcat (Email Analysis) → BlackBird (Username Enumeration) → GhostTrack (Activity Analysis)

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Suspicious URL → Carbon14 (URL Metadata) → WebOSINT (Domain & Infrastructure)



Consolidate Findings → Final Risk Assessment Report

Step-by-Step Workflow:

- 1. Input:** The investigation begins with three core artifacts: a suspicious email, a username, and a URL.
- 2. Email Analysis (Mailcat):** The email address is checked against known data breaches, disposable email services, and its reputation is assessed.
- 3. Username Enumeration (BlackBird):** The username is scanned across hundreds of social media platforms, forums, and websites to build a digital footprint.
- 4. Activity Analysis (GhostTrack):** The username is further analyzed for recent activity, connections, and behavioral patterns on specific platforms.
- 5. URL & Domain Investigation (Carbon14 & WebOSINT):** The provided URL is dissected for technical metadata (Carbon14), while the domain is investigated for registration details, hosting information, and infrastructure (WebOSINT).
- 6. Correlation & Reporting:** Findings from all tools are consolidated to form a comprehensive risk profile of the suspicious identity.

2.2 Tools Overview

For this specific use case, five core tools are employed:

- **BlackBird:** A powerful username investigation tool that searches for a given username across a vast array of social networks.
- **Carbon14:** A URL and metadata analyzer that extracts technical details from a web page, such as technologies used, scripts, and redirections.
- **GhostTrack:** An advanced activity scanner that provides deeper insights into a username's presence and behavior on specific online platforms.
- **Mailcat:** An email verification and breach analysis tool that checks if an email has been compromised in data leaks and assesses its risk level.
- **WebOSINT:** A domain and IP intelligence tool that gathers WHOIS data, hosting provider details, SSL certificate information, and DNS records.

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Tools	Usage
Telescan	Identifies SIM carrier, region, phone metadata
Snoop / Snooper	Scans for suspicious device activity patterns
Ghost Track	Tracks digital footprints, platform activity traces
Yoga	Maps social footprint useful for verifying device user identity
Mailcat	Validates compromised email accounts linked to the device
Blackbird	Username search to match device owner to accounts/logs
Twint-zero	Check if compromised device accessed X/Twitter
Revealin	Examines metadata from files, images, or spyware logs
Sherlog	Log analysis for tracking device-owner inconsistencies
SNWYG	Visual relationship mapping for owners, logs, SIM data
Carbon14	Webpage timestamp verification (useful for phishing links)
Noriarty-Project	Automates multi-tool IR workflows
OSINT-Nap	Executes automated reconnaissance during IR
cloud_enum / cloud_sherlock	Detect cloud leaks or backup exposure
SonicVisualiser	Analyzes suspicious audio recordings (spy apps, calls)
Unredacter	Used only for testing pixelation reversal (ethical limits)
ExchangeFinder	Checks if device communicated with corporate Exchange server
Fake-SMS	For testing SMS spoofing scenarios (non-operational)
WebOSINT	Open-source web footprint investigation
Zen	Reporting and OSINT workflow optimization

3. Features and Functionalities

The OSINT machine provides the following key features:

Centralized Forensic Workstation: All tools are pre-installed and configured in a single, isolated VM, eliminating setup overhead and ensuring a consistent analysis environment.

Automated Data Collection: Automates the tedious process of querying multiple OSINT sources manually, significantly reducing investigation time.

Comprehensive Identity Profiling: Builds a multi-faceted profile of a digital identity by correlating data from email, username, and web-based sources.

Phishing & Scam Detection: Specifically designed to identify hallmarks of phishing campaigns, such as newly registered domains, disposable emails, and impersonation accounts.

Evidence Preservation: Allows for the systematic saving of tool outputs within the VM, creating an auditable trail for the investigation.

Risk Scoring: Tools like Mailcat provide direct risk scores, while other outputs allow the analyst to assign an overall threat level.

4. Cyber Law and Ethics Consideration

Operating within legal and ethical boundaries is paramount for any OSINT activity.

4.1 Legal Constraints

Terms of Service (ToS): We strictly use tools that aggregate publicly available information. We do not automate scraping of platforms that explicitly forbid it in their ToS. The tools themselves (BlackBird, etc.) operate by accessing public profiles and data, which is generally permissible.

Data Privacy Laws (GDPR, CCPA, etc.): Our investigation is targeted at a suspicious entity potentially involved in malicious activity, falling under the "legitimate interest" clause for investigation. We only collect and process data that is already publicly accessible. We do not seek out or store special categories of personal data.

Computer Fraud and Abuse Act (CFAA) & Similar Laws: All interactions with external systems are conducted in a manner that does not constitute unauthorized access. We only request and receive publicly available data without bypassing any access controls.

4.2 Ethical Considerations

Purpose Limitation: The machine is designed for cybersecurity threat investigation and educational purposes. It is not intended for harassment, stalking, or unauthorized corporate espionage.

Minimization of Data Collection: We only collect data directly relevant to the suspicious identity under investigation. The machine does not perform broad, untargeted data collection.

Responsible Disclosure: If our investigation were to uncover a critical vulnerability or a large-scale data breach on a legitimate platform, our policy would be to responsibly disclose this information to the affected organization.

Transparency: This report and the associated documentation clearly state the capabilities and intended use of the OSINT machine to prevent misuse.

5. Challenges Faced and Solutions

Challenge 1: Tool Compatibility and Dependencies.

Description: Many OSINT tools are written in different languages (Python, Go, Rust) and have complex, often conflicting, dependencies.

Solution: By using a virtual machine, we created a sandboxed environment where each tool's dependencies could be installed and managed without affecting the host system or other tools. Docker containers were considered for some tools for even greater isolation.

Challenge 2: Information Overload and Correlation.

Description: Each tool generates a significant amount of data. Manually correlating findings from five different tools to form a coherent narrative can be challenging.

Solution: We designed a standardized workflow and report structure. Future improvements include developing a master script or dashboard to automatically parse and correlate key findings into a single report.

Challenge 3: Maintaining Tool Efficacy.

Description: OSINT tools can break frequently as the websites and APIs they rely on change.

Solution: We implemented a maintenance routine to periodically update the tools within the VM. The modular architecture also allows for easy replacement of a tool if it becomes obsolete.

6. Case Study Demonstration:

Scenario:

A reported email primefundshelpdesk@gmail.com with username **primefunds_support** and URL <https://primefunds.com/verify> is investigated using the OSINT VM to verify legitimacy and ensure it is safe.

Investigation Steps:

1. **Mailcat:** Checked for past data breaches; no breaches detected, email flagged as low-risk.
2. **BlackBird:** Username found on professional forums; no scam or phishing association.
3. **GhostTrack :** Activity patterns indicate normal usage; no signs of impersonation.
4. **Carbon14:** URL analyzed; no phishing scripts detected, metadata consistent with legitimate business operations.
5. **WebOSINT:** Domain registered for a significant period, hosted in a trusted region, SSL certificate active.

Final Analysis:

The digital identity appears legitimate, with no indicators of phishing or scam operations. The OSINT investigation confirms that communications from this email and domain are safe. Cyber teams can consider this verified and continue normal operations.

7. Conclusion and Potential Improvements

We have successfully designed and implemented a functional, specialized OSINT machine capable of conducting a thorough investigation of a suspicious digital identity. The integrated use of BlackBird, Carbon14, GhostTrack, Mailcat, and WebOSINT provides cybersecurity professionals with a powerful, rapid-response capability to assess threats like phishing and impersonation.

Potential Improvements:

Automated Reporting Module: Develop a Python script that runs all five tools with a single command, parses their output, and generates a unified PDF or HTML report.

Graphical User Interface (GUI): Create a simple web-based dashboard inside the VM to launch tools and view results, making it more accessible to less technical users.

Threat Intelligence Integration: Incorporate APIs from threat intelligence platforms (e.g., VirusTotal, AbuseIPDB) to enrich the data further.

Extended Toolset: Create modular "profiles" for different use cases that can enable or disable different sets of tools from the main VM repository.

8. References

- The tools used are open-source and their repositories can be found on GitHub.
- General Data Protection Regulation (GDPR)
- California Consumer Privacy Act (CCPA)
- Computer Fraud and Abuse Act (CFAA)